

Normalization

1. Observe the structure of a given EMPLOYEE table as below and suggest if this table can be further Normalized. EMPNO, ENAME, SAL, DEPTNO, DNAME, LOC Explain what normal form this table is and how you can make this into the next normal form.

Given Table:

EMPNO	ENAME	SAL	DEPTNO	DNAME	LOC
101	Ravi	50000	10	HR	Hyderabad
102	Sita	45000	20	IT	Bangalore
103	Amit	55000	10	HR	Hyderabad

Primary Key: EMPNO

Observation: DNAME and LOC are repeated for every employee in the same department.

Functional Dependencies:

EMPNO → ENAME, SAL, DEPTNO

DEPTNO → DNAME, LOC

Current Normal Form: 2NF

Reason: Since DNAME and LOC depend on DEPTNO instead of EMPNO, a transitive dependency exists. Hence the table is not in 3NF.

Convert into 3NF:

EMPLOYEE

EMPNO	ENAME	SAL	DEPTNO
101	Ravi	50000	10
102	Sita	45000	20
103	Amit	55000	10

DEPARTMENT

DEPTNO	DNAME	LOC
10	HR	Hyderabad
20	IT	Bangalore

2. Observe the structure of a given STUDENT table as below and suggest if this table can be further Normalized.ROLLNO, NAME, AGE, EXAM, MARKS, GRADE.Explain what normal form this table is and how you can make this into the next normal form.

Given Table:

ROLLNO	NAME	AGE	EXAM	MARKS	GRADE
1	Rahul	20	DBMS	85	A
1	Rahul	20	Java	78	B
2	Priya	21	DBMS	92	A+

Composite Primary Key: (ROLLNO, EXAM)

Observation: NAME and AGE repeat for every exam written by the same student.

Functional Dependencies:

ROLLNO → NAME, AGE

(ROLLNO, EXAM) → MARKS

MARKS → GRADE

Current Normal Form: 1NF

Reason: Partial dependency (ROLLNO → NAME, AGE) and transitive dependency (MARKS → GRADE).

Convert into 3NF:

STUDENT

ROLLNO	NAME	AGE
1	Rahul	20
2	Priya	21

EXAM_RESULT

ROLLNO	EXAM	MARKS
1	DBMS	85
1	Java	78
2	DBMS	92

GRADE

MARKS	GRADE
85	A
78	B
92	A+

3. Observe the structure of a given EMPLOYEE table and suggest what kind of problem this table has and how to solve the problem.EMPNO, PROJECT_NO, NO_OF_DAYS, CUSTOMERNAME.In the above table, EMPNO and PROJECT_NO both together are a composite primary key.

Given Table:

EMPNO	PROJECT_NO	NO_OF_DAYS	CUSTOMERNAME
101	P101	20	ABC Ltd
101	P102	15	XYZ Ltd
102	P101	10	ABC Ltd

Composite Primary Key: (EMPNO, PROJECT_NO)

Observation: CUSTOMERNAME repeats whenever the same PROJECT_NO appears.

Functional Dependencies:

(EMPNO, PROJECT_NO) → NO_OF_DAYS

PROJECT_NO → CUSTOMERNAME

Current Normal Form: 1NF

Reason: CUSTOMERNAME depends only on PROJECT_NO (partial dependency), so the table is not in 2NF.

Convert into 2NF:

EMPLOYEE_PROJECT

EMPNO	PROJECT_NO	NO_OF_DAYS
101	P101	20
101	P102	15
102	P101	10

PROJECT

PROJECT_NO	CUSTOMERNAME
P101	ABC Ltd
P102	XYZ Ltd